

Esthetics and Operative Dentistry

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INTRODUCTION

- ▶ Def. What is esthetics?
- ▶ It is pertaining to a sense of the beautiful.
- ▶ It is an art where scientific principles are applied for creating aesthetics to create restorations and prosthesis, dental aesthetics also depends upon mathematical concepts.

- ▶ The main aim of operative care is achieved by attaining health, function and aesthetics. It is also known as health, function and aesthetic triad (HFA).
- ▶ The sequence of this triad is achievement of health, followed by function and then by aesthetics.

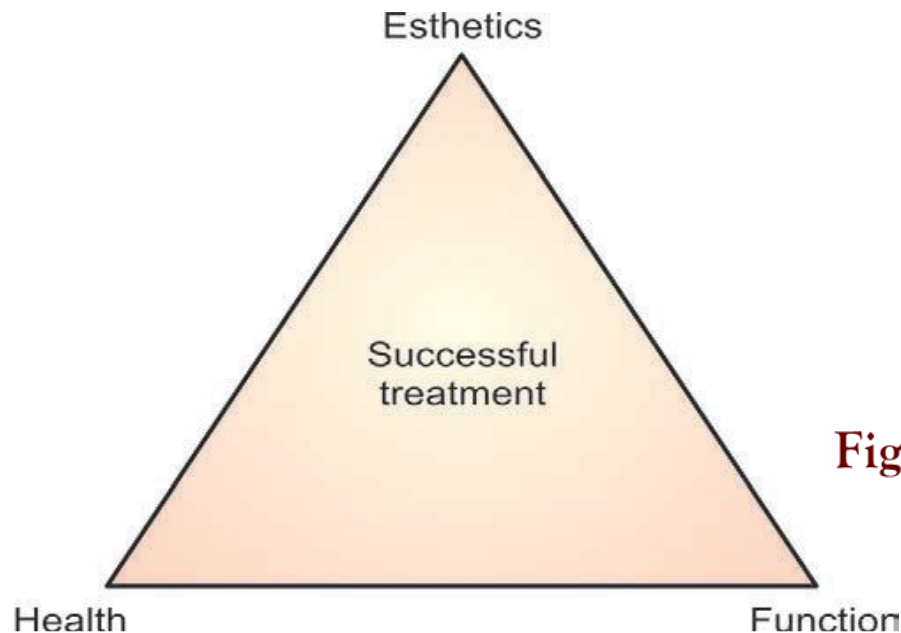


Fig:1. HFA triad for achieving optimal treatment

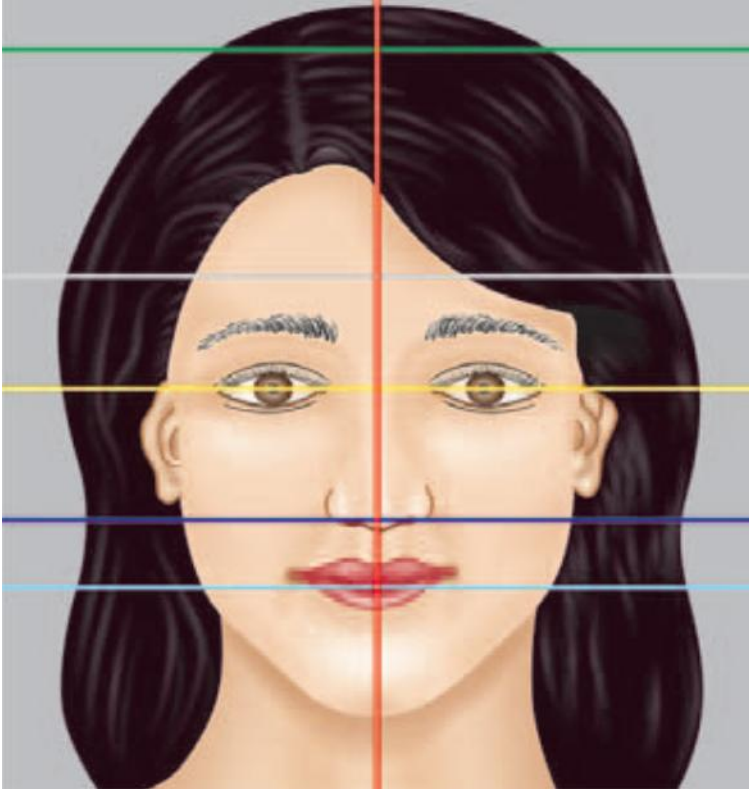
DENTAL AESTHETICS

Dental esthetics can be divided into four components

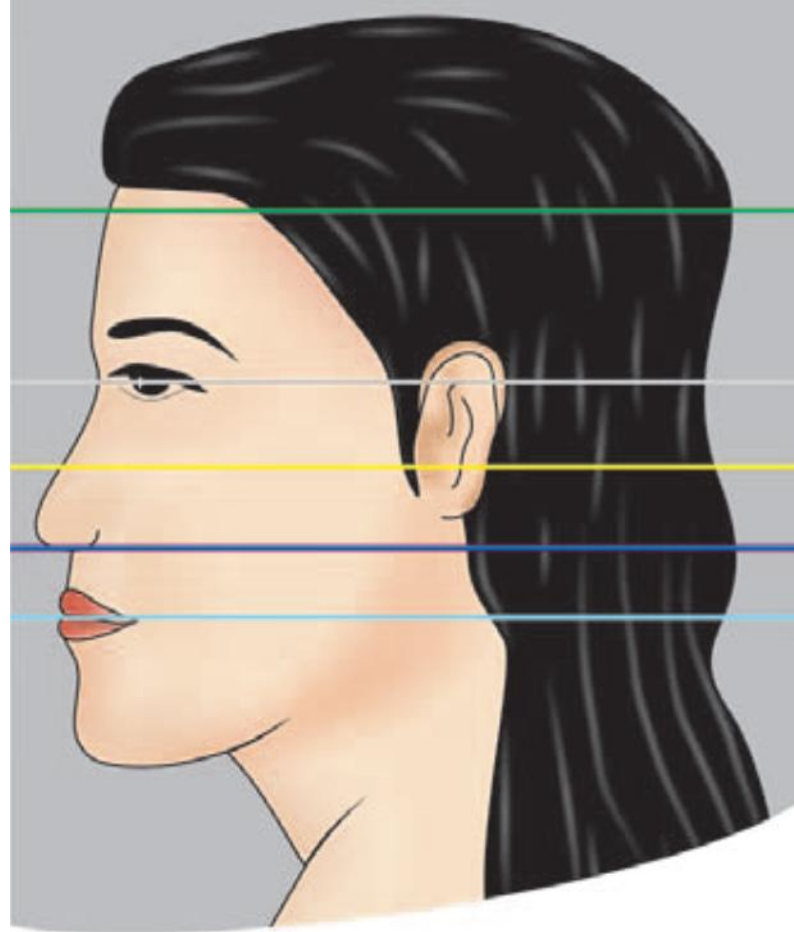
- ▶ 1. Facial composition
- ▶ 2. Dentofacial composition
- ▶ 3. Dental composition
- ▶ 4. Gingival composition.

1. FACIAL COMPOSITION

- ▶ Facial composition is measured by analyzing the face from frontal and sagittal aspect.
 - a. From frontal aspect:
 - ▶ Many landmarks are used to determine aesthetics.
 - ▶ Many horizontal lines are drawn horizontally from upper to lower part of face, these are hair ophriac, interpupillary, interalar and commissural lines.
 - ▶ Parallelism of these lines result in horizontal symmetry of the face.



Frontal aspect of facial
composition



Parallelism of facial lines makes
a face symmetrical

b. From sagittal aspect:

- ▶ Two reference points are used that is nasolabial angle and Rickett's E plane.
- ▶ Nasolabial angle: - is formed by intersection of two lines using nose and lips as reference points (Fig.4.4).
- ▶ Normal value of Nasolabial angle for males is 90° - 100° and for females is 100° - 105° .
- ▶ Rickett's E-plane: is line drawn from tip of nose to chin prominence (Fig.4.5).
- ▶ Using these two reference points, protrusion or retrusion of maxilla can be evaluated.

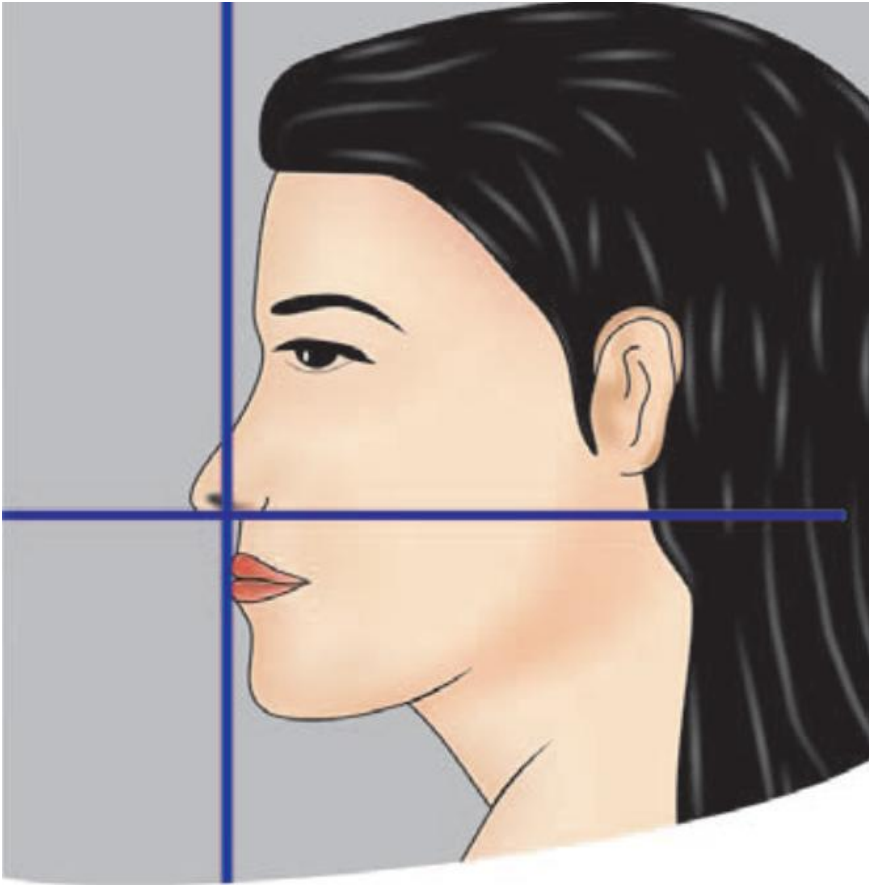


Fig. 4.4: Nasolabial angle is formed by intersection of two lines using nose and lips as reference points

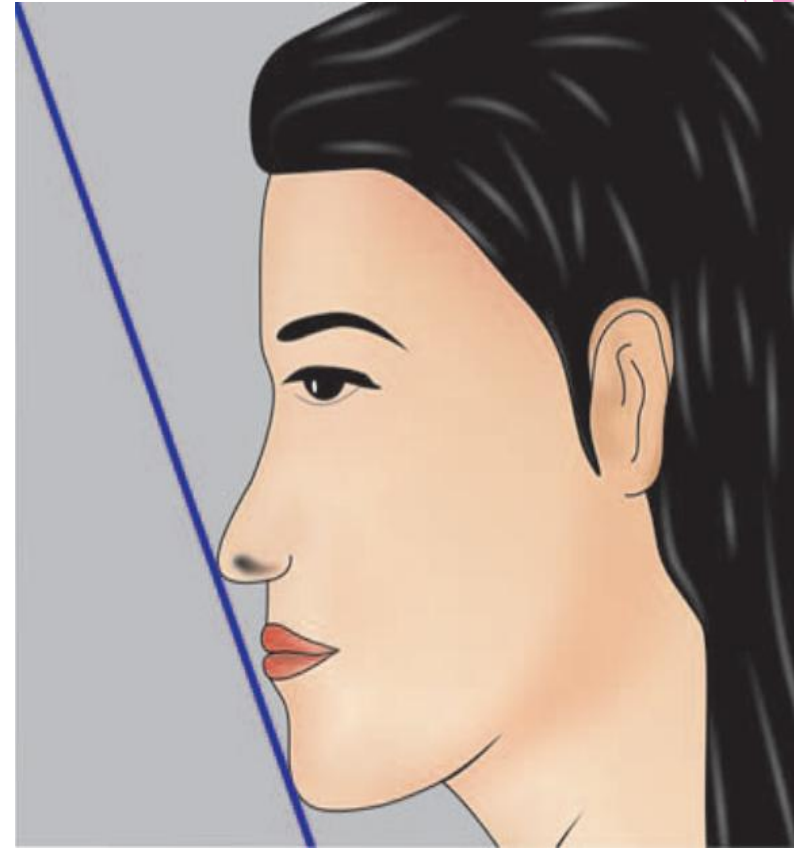


Fig.4.5: Ricket's E-plane is line drawn from tip of nose to chin prominence

2. DENTOFACIAL COMPOSITION

- ▶ Dentofacial principles include evaluation of the collective appearance of the teeth in relation to the lips. It is determined by two positions:
 - a. Static
 - b. Dynamic
- ▶ **Static**
- ▶ In static position, teeth are out of occlusion, and tips are slightly parted. This position are affected by lip length, age and sex of the individual.

- ▶ **Dynamic**
- ▶ This position is determined by smile.
- ▶ Depending on facial musculature, shape and size of lips, skeletal make up, the extent of tooth exposure occurring during smile varies.
- ▶ **A smile line:** is an imaginary line which is drawn on the **tips of maxillary teeth parallel to curvature of lower lip** (Fig.4.6).
- ▶ **A good smile:** is that when upper anterior teeth are in line with the curvature of lower lip, and the corners of lips are raised to same height on both sides.

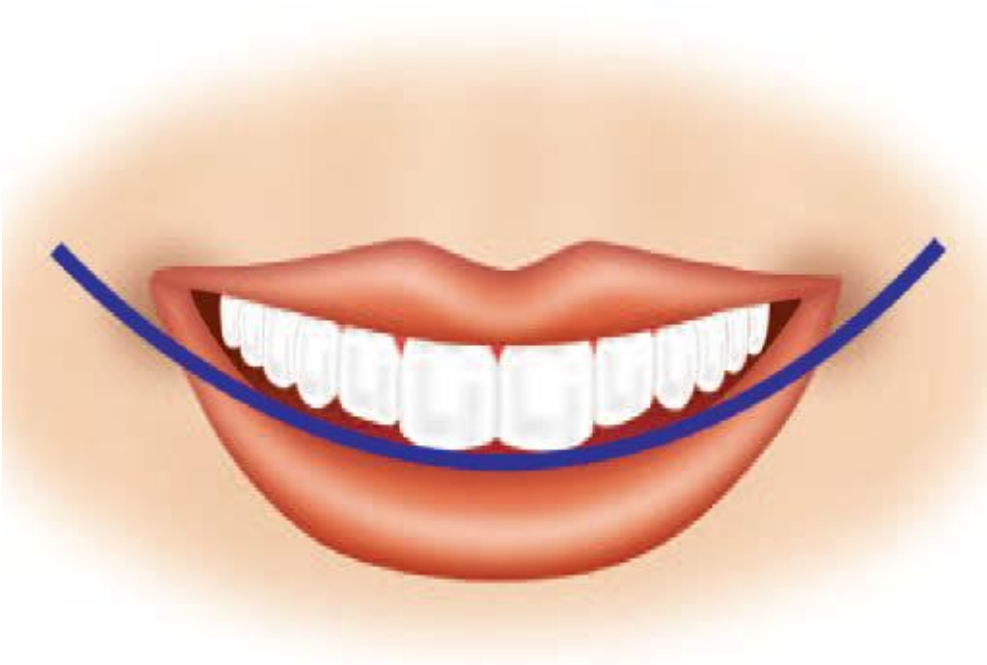
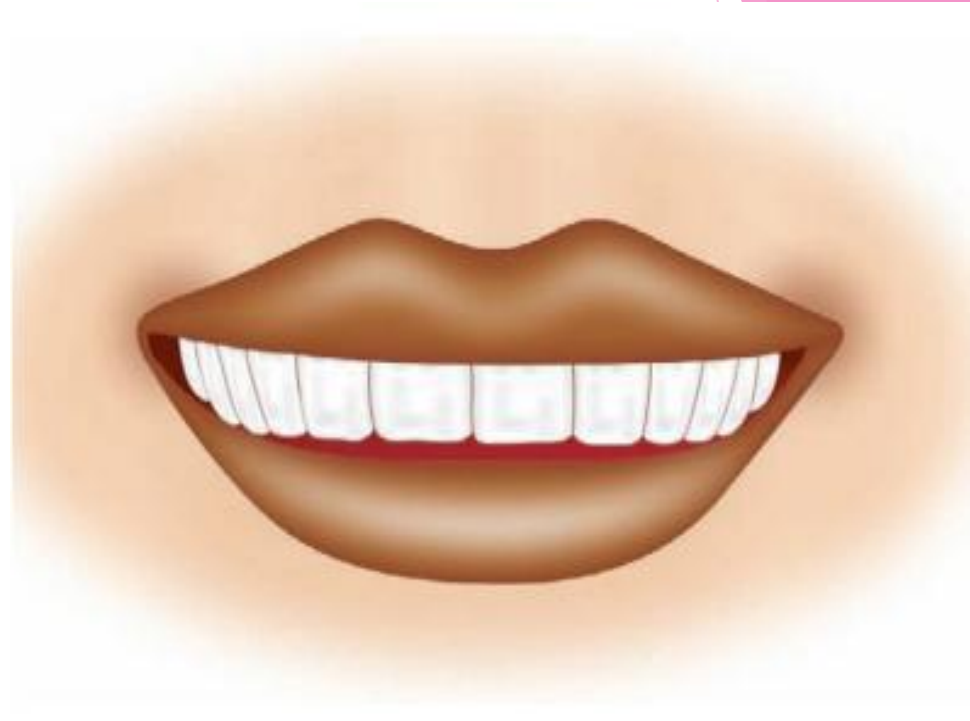


Fig.4.6: Smile line and feminine smile shows teeth with rounded incisal angles, open incisal and facial embrasures.



Masculine smile shows close incisal embrasures with prominent incisal angles

- ▶ Dental aesthetics comprises the touch of contour and color characteristics of individual teeth.
- ▶ The contour of a tooth can be analyzed in three-dimensional view: from the facial, interproximal, and incisal.
- ▶ The “face” of a tooth has been defined as the reflective area inside the transitional line angles of the facial surface.
- ▶ It consists of size and shape of teeth along with their interarch and intra-arch relationship.

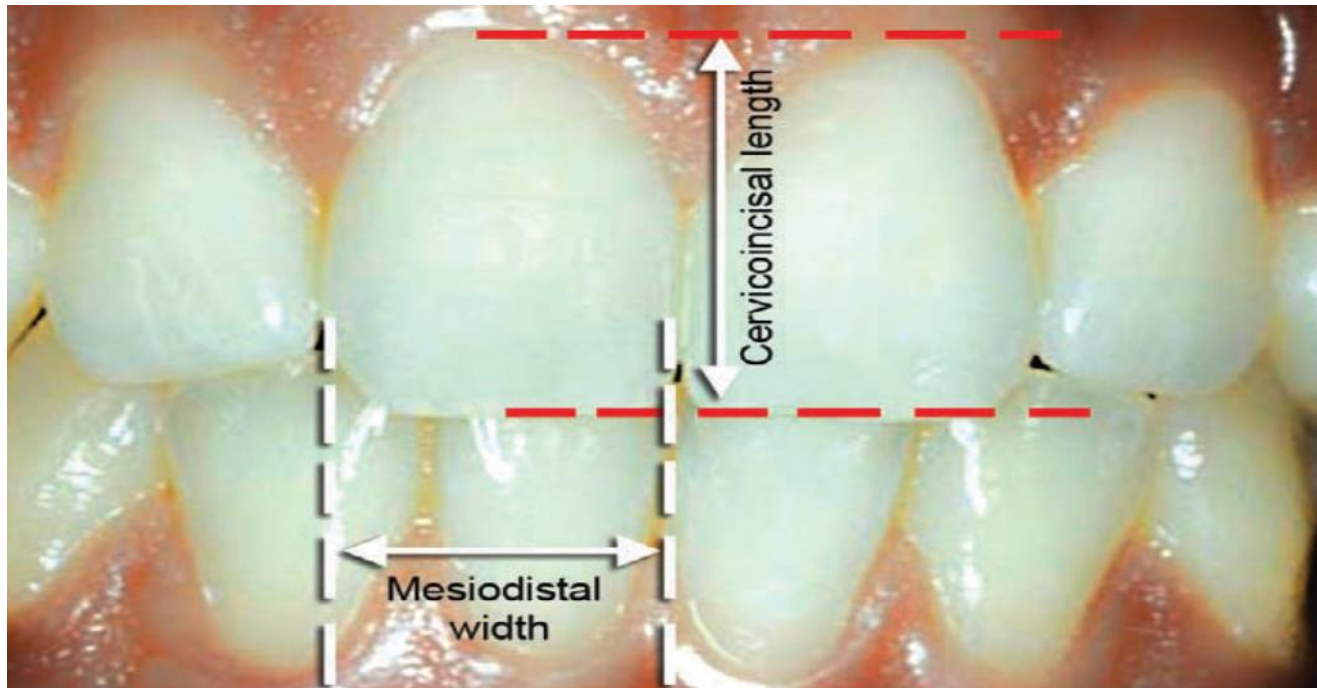
3. DENTAL COMPOSITION

▶ ARTISTIC ELEMENTS

- ▶ Best conservative esthetic results can be obtained by the consideration of the following factors:
 - ▶ 1. Shape or form
 - ▶ 2. Symmetry and proportionality
 - ▶ 3. Position and alignment
 - ▶ 4. Surface texture
 - ▶ 5. Color
 - ▶ 6. Translucency
 - ▶ 7. Defect and damage repair.

1. Size of tooth (SOT):

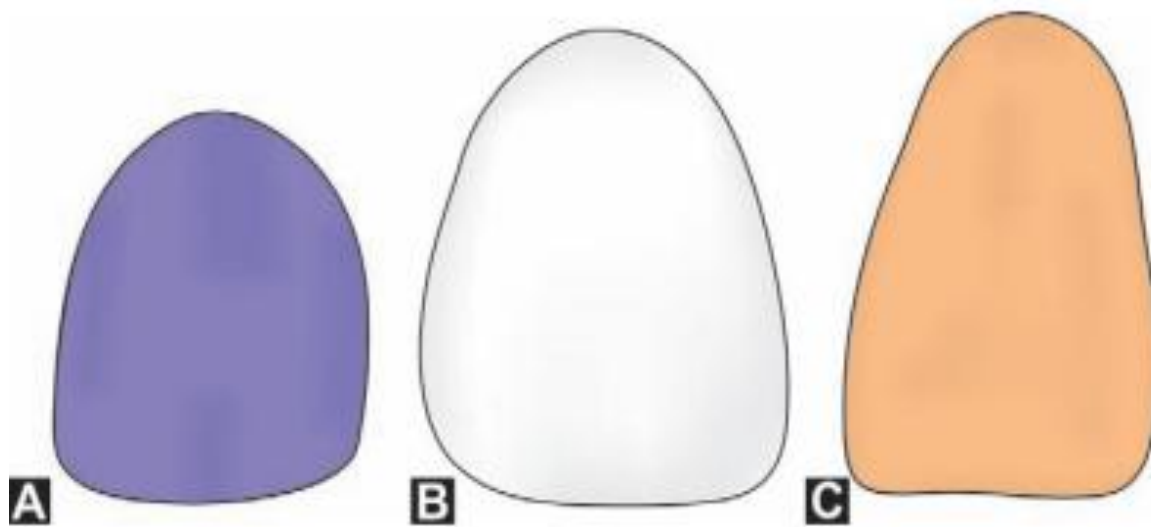
- ▶ It is determined by dividing cervicoincisal length of tooth to mesiodistal width, i.e.
- ▶ Size of tooth = Width/length ratio.



Size of tooth is determined by dividing cervicoincisal length to mesiodistal width

► To have optimal dimension, width/length ratio of central incisor should range from 0.75 - 0.8 .

- • Ideal ratio 0.75-0.8
- • < 0.75 - Narrower tooth
- • > 0.8 - Wider tooth



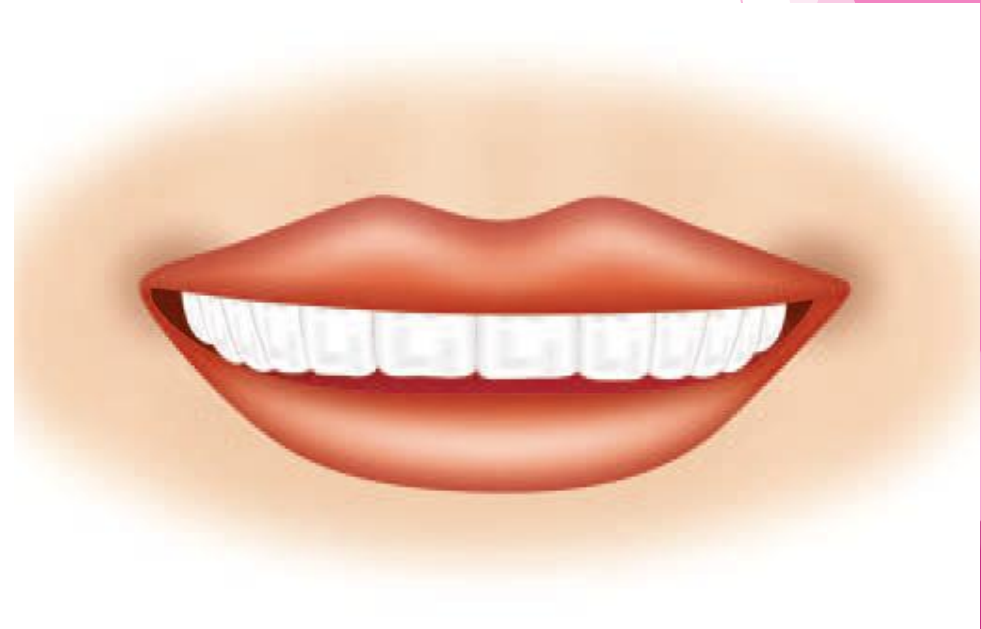
Width/Length ratio: (A) > 0.8 wider tooth; (B) 0.75-0.8 Normal tooth; (C) ≤ 0.75 Narrower tooth

2. Shape of upper anterior teeth

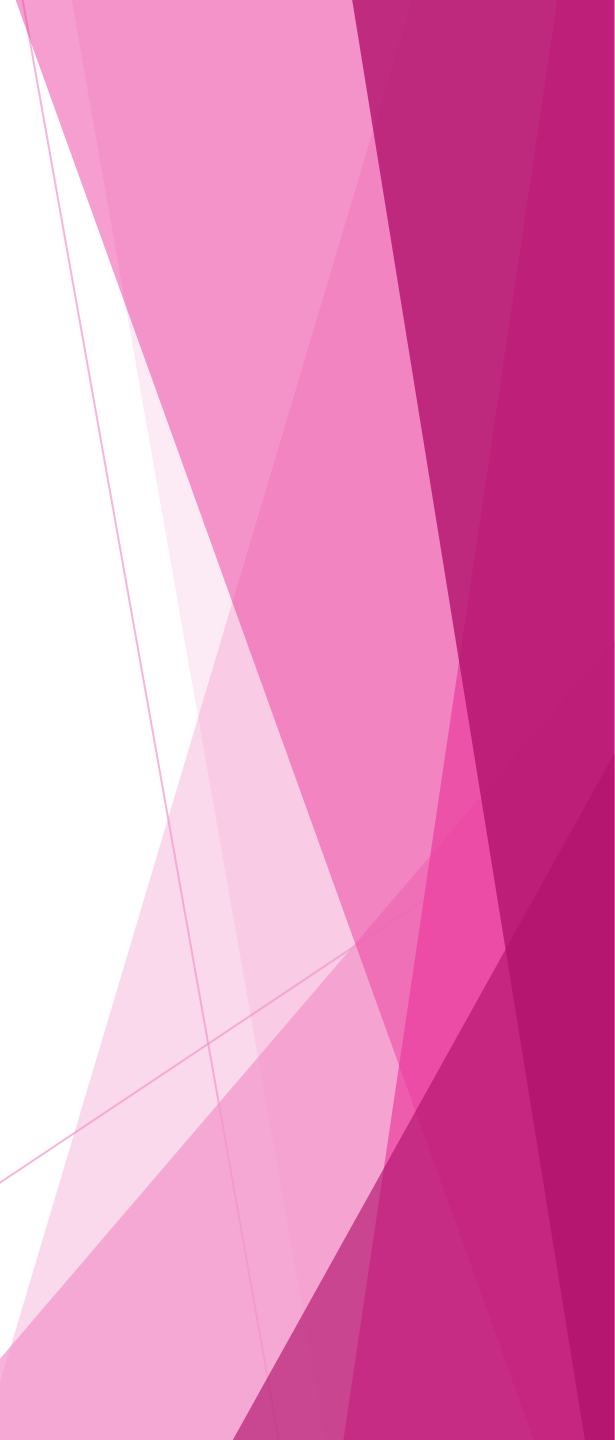
- ▶ It is determined by age, sex and personality of the individual.
- ▶ For example, a young and feminine smile shows: teeth with rounded incisal angles, open incisal and facial embrasure while a masculine smile shows: closed incisal embrasures with prominent incisal angles .



Youthful feminine teeth have rounded incisal angles, open incisal and facial embrasures

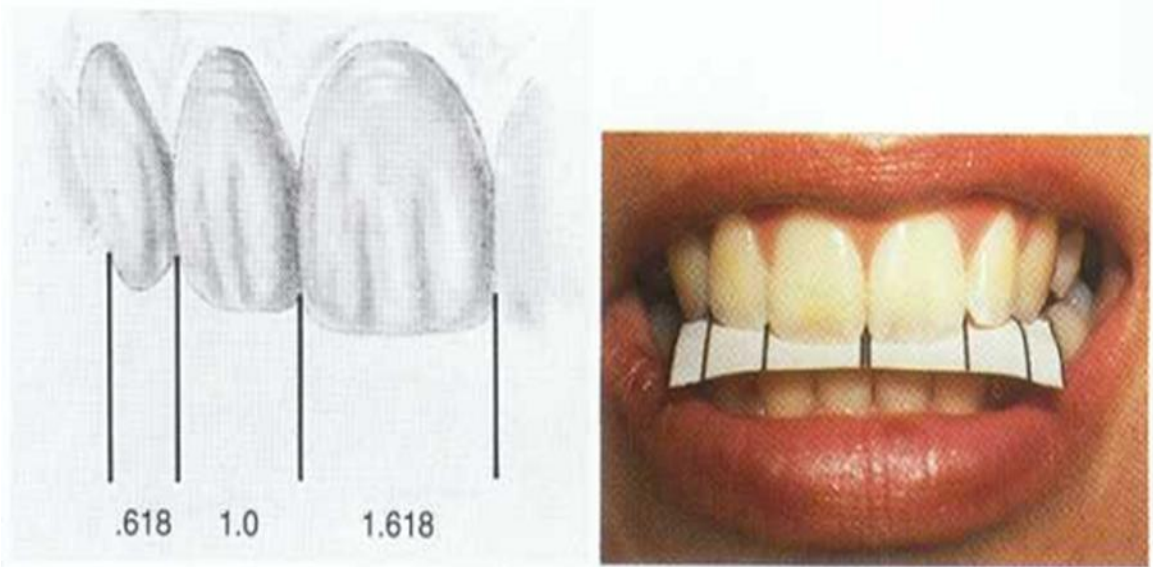


Masculine teeth show closed incisal embrasures and prominent incisal angles

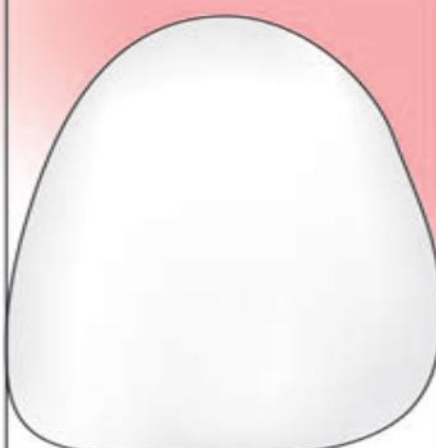
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- ▶ **NOTE:**
 - ▶ If in females slightly **broader teeth** are present, then they sometimes require **conservative minor modification to produce better aesthetics.**
 - ▶ This is called '**cosmetic contouring**'. In this to create younger and more feminine smile, **incisal angles are rounded and incisal embrasures are opened.**

3. Symmetry and proportionality

- ▶ A. Golden proportion
- ▶ This concept was given by Lombardi and Levin. According to it, for an object to be proportional to one another the ratio of 1:1.618 is aesthetically pleasing.
- ▶ Based on this, when viewed from the front, is considered to be esthetically pleasing if each tooth in that smile (starting from the midline) is approximately 60% of the size of the tooth immediately mesial to it.
- ▶ The exact proportion of the smaller tooth to the larger tooth is 0.618.
- ▶ It must be emphasized that these proportions are based on the apparent sizes of the teeth when viewed straight on and not the actual sizes of the individual teeth.



A, The exact ratios of proportionality. B, The anterior teeth of this patient are in "golden proportion" to one another

	Golden proportion		
			
Ratio	1.618	1	0.618
Apparent width	9 mm	$9 \text{ mm} \times 0.618 = 5.6 \text{ mm}$	$5.56 \times 0.618 = 3.5 \text{ mm}$

- ▶ Recurring Esthetic Dental Proportion (Red Proportion)
- ▶ RED proportion states that the width of the teeth as viewed from frontal should remain constant as one move distally instead of using 62 % proportion.

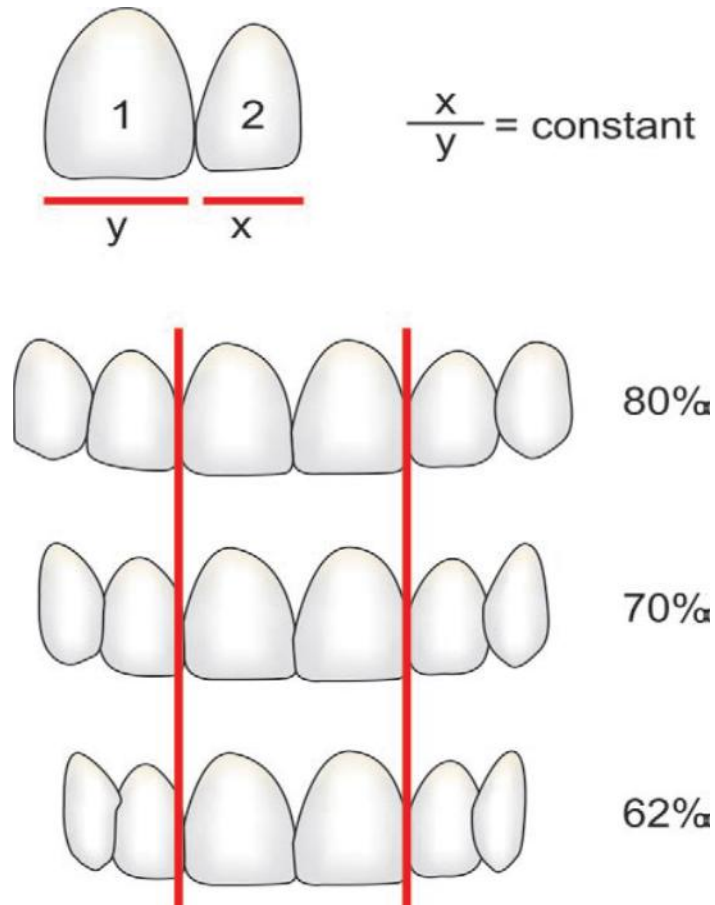
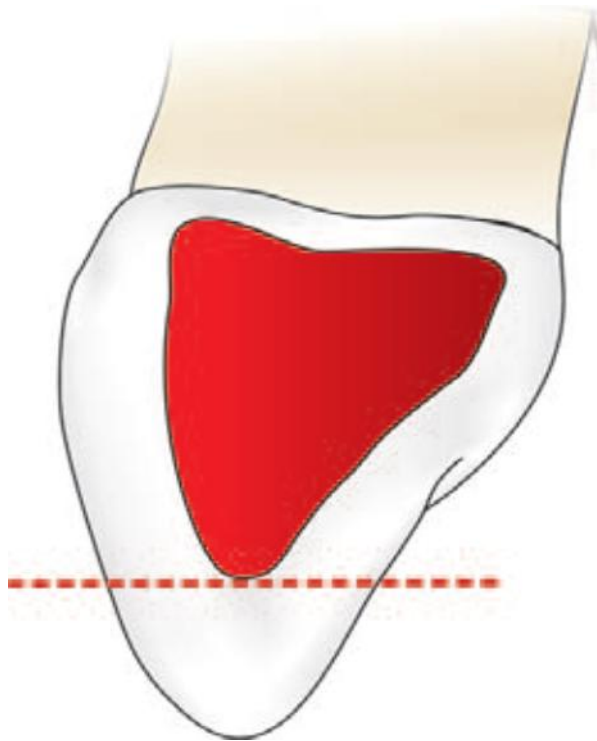


Figure: RED proportion

- ▶ b. Axial inclination of the tooth:
- ▶ The maxillary anterior teeth ideally display mesial axial inclination, with the central incisors appearing to be the almost vertical and the lateral incisors and canines each tipping more toward the midline.
- ▶ After the canines, the posterior teeth display an inclination that is parallel to the canines.

- ▶ c. Labiolingual thickness of anterior teeth
- ▶ It is measured at the junction of middle and incisor third of tooth.
- ▶ Ideally it should be between 2.5-3.5 mm.
- ▶
 - If > 3.5 mm, it is over contouring.
 - If < 2.5 mm, it necessitates aesthetic procedures.



Labiolingual thickness of anterior teeth is measured at junction of middle and incisor third of the tooth

4. Proper position and alignment

- ▶ Are necessary for good appearance of face and smile.
- ▶ Major defects in position and alignment are corrected by orthodontic Rx, but minor positional defects can be treated by composites or facial veneers.

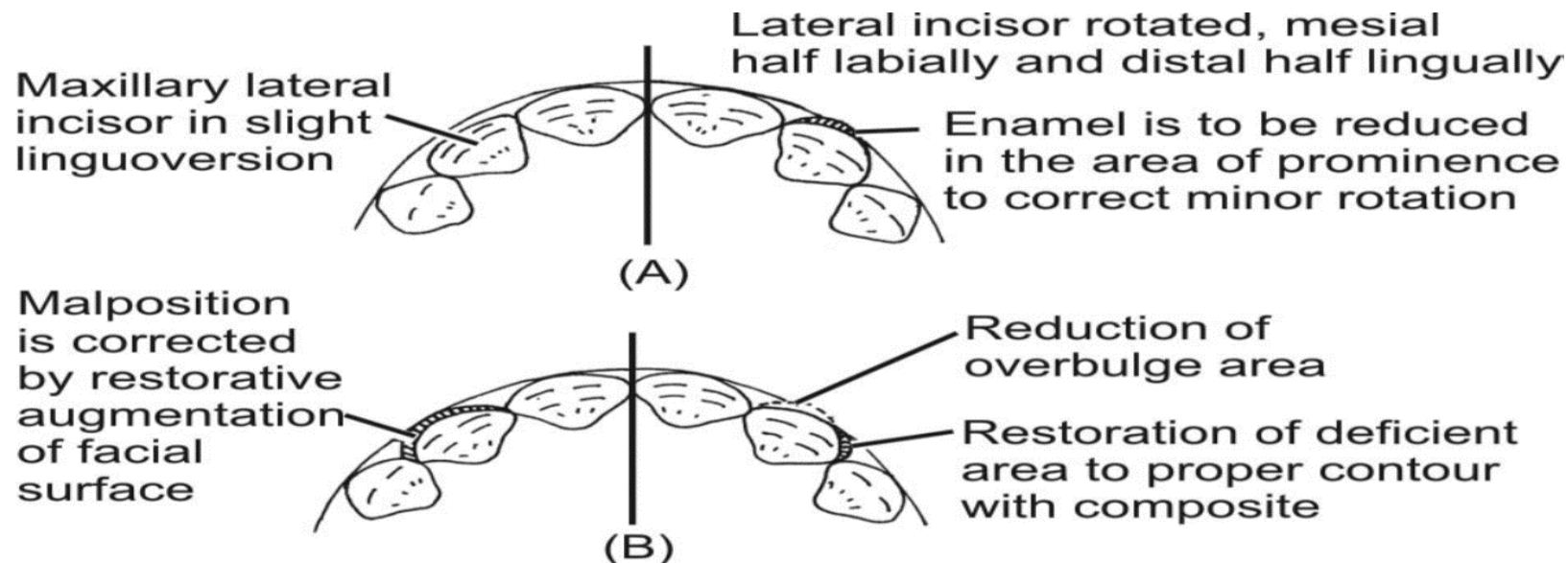


Fig: Correction of alignment and position with restoration and reduction in anterior teeth. (A) Before correction, (B) After correction

5. Surface Texture

- ▶ Teeth of young persons show various surface characterizations but in the older person, teeth appear smooth and lack surface characterizations.
- ▶ Close examination of various anatomical structures like facets, cervical perikymata, developmental depression on the natural tooth which reflect light in many directions helps very much to create same characters in the restored part of the tooth.

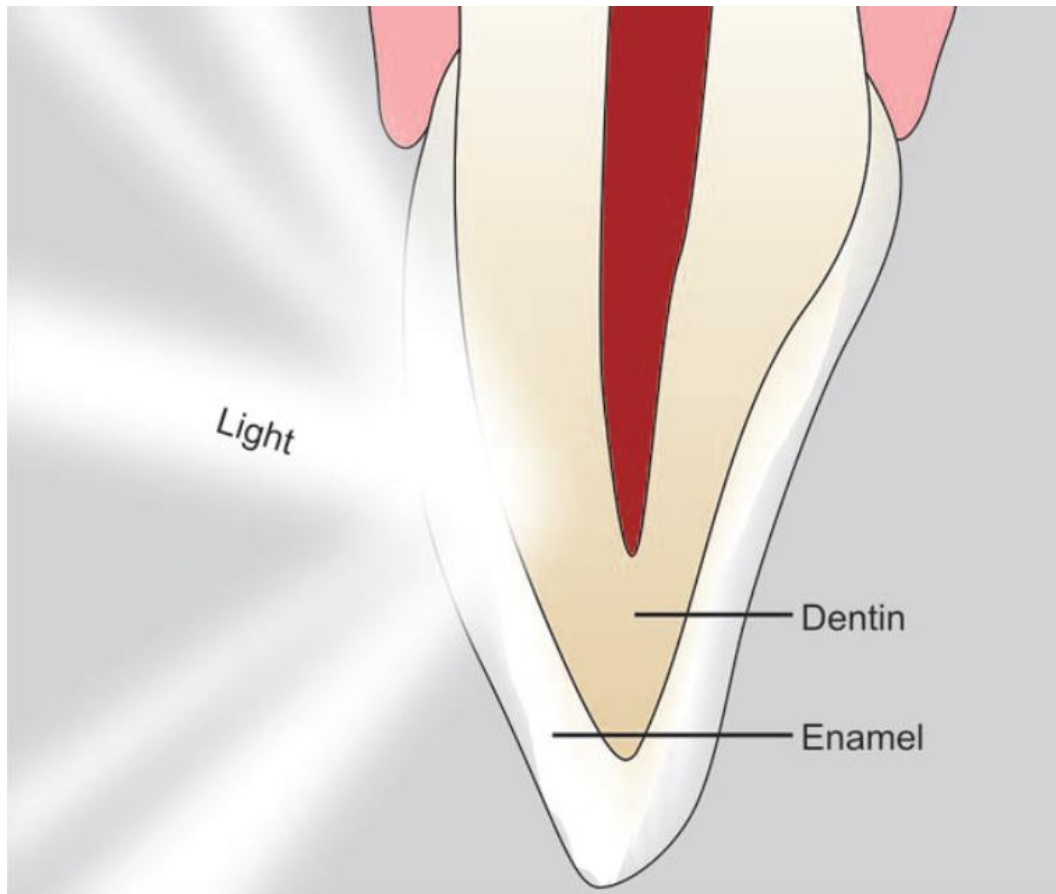
6. Color

- ▶ Color has the most important and very complex role in conservative dentistry.
- ▶ For example:
 - ▶ 1. The gingival third has darker shade and as we move incisally, the shade becomes lighter. This means the restoration of a single tooth may require different shades in different areas so as to have optimal results.
 - ▶ 2. In young persons, thick enamel is present making the shade of teeth lighter as compared to old persons, which have darker shade.
 - ▶ 3. In old persons, the wear of enamel takes place and the dark color of the dentin becomes prominent through thin enamel giving darker shade.

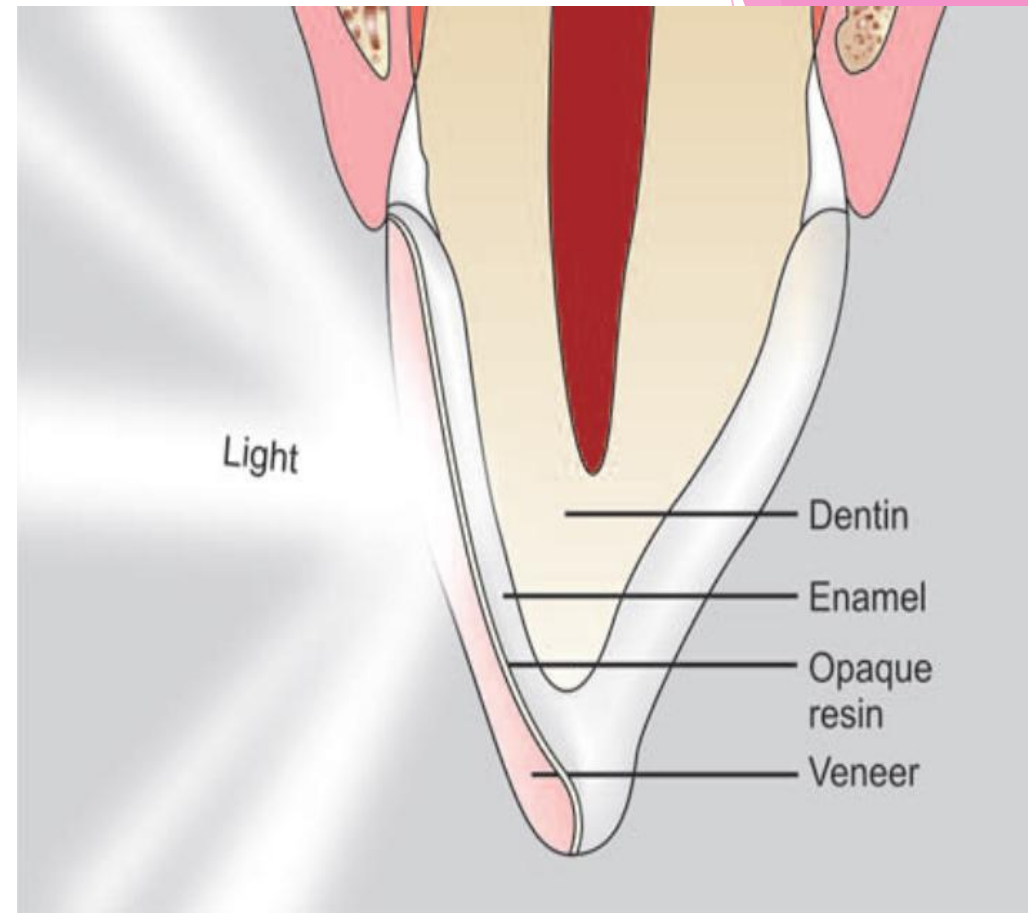
- ▶ Thus, a proper knowledge of normal tooth color enables a dental surgeon to create a restoration that appears natural.
- ▶ Color of the teeth appears different in different lights. This phenomenon is called 'metamerism'.
- ▶ Shade selection should be done in daylight. Shade selection should be done before isolation of tooth. It prevents color changes due to drying and dehydration of teeth.
- ▶ During shade selection, the eyes get fatigued after few seconds so eyes should be relaxed by seeing at a blue object.

7. Translucency

- ▶ has played an important role in aesthetics and in creation of natural appearance.
- ▶ In natural healthy teeth, the light penetrates the enamel and goes a little bit deeper into the dentin, after this it is reflected giving life like appearance.
- ▶ However in case of restoration either metal, composite resin or porcelain, the light is reflected from the outer surface. Hence, aesthetic appearance of vitality is compromised.



In normal tooth, light deeply penetrates into enamel and dentin before being reflected outwards



In case of restored tooth, presence of opaque media limits the penetration of light giving less aesthetic look

4. GINGIVAL COMPOSITION

- ▶ The cementoenamel junction and the osseous crest determine the curvature of the gingival margin of the tooth.
- ▶ This curvature of the gingival margin of the tooth is called gingival shape.
- ▶ The mandibular incisors and the maxillary laterals should display a gingival shape of symmetrical half oval or half - circular shape and that of the maxillary centrals and canines a gingival shape that is more elliptical

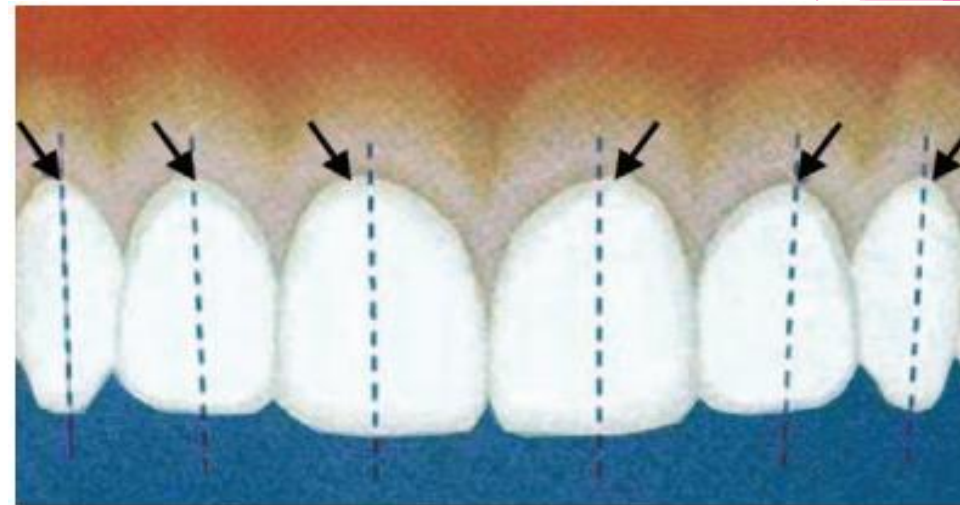


Gingival shape of incisors and canine

- ▶ The gingival zenith represents the most apical point at which each tooth emerges from the free gingival margin.
- ▶ For an esthetically pleasing smile, it should be positioned distally to the center of each tooth within the maxillary anterior segment. The distal position gradually increases from the central incisor to the canine
- ▶ The gingival zenith of the maxillary laterals and mandibular incisors should coincide with their longitudinal axis.



Gingival zenith is the most apical point at which each tooth emerges from free gingival margin



Zenith point should be located distal to the center of each tooth. Distal position gradually increases from the central incisor to the canine

Conservative Esthetic Procedures

- ▶ Many treatment options are available which can be employed to improve the aesthetics of affected tooth/ teeth. There are number of problems which can alter the aesthetics of anterior teeth.
- ▶ Commonly seen problems with anterior teeth which affect aesthetics of these teeth are:
 - A. Caries
 - B. Tooth discoloration because of trauma, hypoplasia and other factors
 - C. Tooth malformations
 - D. Diastema between teeth
 - E. Malalignment of teeth
 - F. Fracture of tooth
 - G. Cervical lesions like erosion, abrasion and abfraction
 - H. Attrition of teeth
 - I. Ectopic eruptions.

Treatment

- ▶ Following treatment options are available for correction of these conditions. More than one option can be advocated in some cases so as to achieve optimal results:
- ▶ **Procedures:**
 1. Ameloplasty/enameloplasty
 2. Bleaching of teeth
 3. Restorations with composite resins
 4. Orthodontic treatment
 5. Veneering:
 - Composite
 - Porcelain
 - Metal ceramic
 6. Full coverage crown.

1. Ameloplasty/Enameloplasty

- ▶ It helps in improving minor changes in contour of tooth by removal of enamel.
- ▶ It is commonly used to smoothen the roughened enamel margins, fractured tooth surfaces and to soften interproximal angles.
- ▶ Tooth is polished after ameloplasty so as to have fine scratch proof appearance.

3. Restorations with Composite Resins

- ▶ Composite resins are indicated for treatment of minor defects present on incisal edges or labial surfaces of teeth like caries, fracture of teeth and for the modification of anatomy and morphology of teeth like correction of diastema, peg shaped laterals, etc.
- ▶ Composite resins have advantages of being aesthetic, noninvasive, inexpensive, simple to use but they tend to discolor and wear with time.

► Technique of Using Composite Resins

- Do conservative tooth preparation or just roughen the enamel surfaces.
- Do etching, rinsing and drying of the tooth.
- Apply dentin bonding agent and cure it.
- Apply composite of selected shade to the prepared tooth surface and cure it
- Finally do finishing and polishing of the restoration.

4. Orthodontic Treatment

- ▶ Orthodontic treatment is done to improve functional relationship of arches and to align the teeth properly for enhancing aesthetics.
- ▶ Orthodontics treatment is usually opted for correction of malaligned teeth, diastema, ectopic eruptions especially when patient is young and dentition is more amenable to rearrangement.

5. Veneers

- ▶ Veneer can be described as a layer of tooth colored material which is applied on the tooth surface for aesthetic purpose.
- ▶ They are used to mask the localized, generalized defects and intrinsic discolorations.
- ▶ Indications:
 - Damaged, defective and malformed facial surface
 - Discolored facial surface
 - Discolored restorations.

Types of Veneers

▶ 1. Based on method of fabrication:

- Direct technique
- Indirect technique

▶ 2. Based on extent of coverage:

• Partial veneers:

- are used for the localized damage, defect and discoloration of the tooth, i.e. they involve only a portion of the tooth crown

• Full veneers:

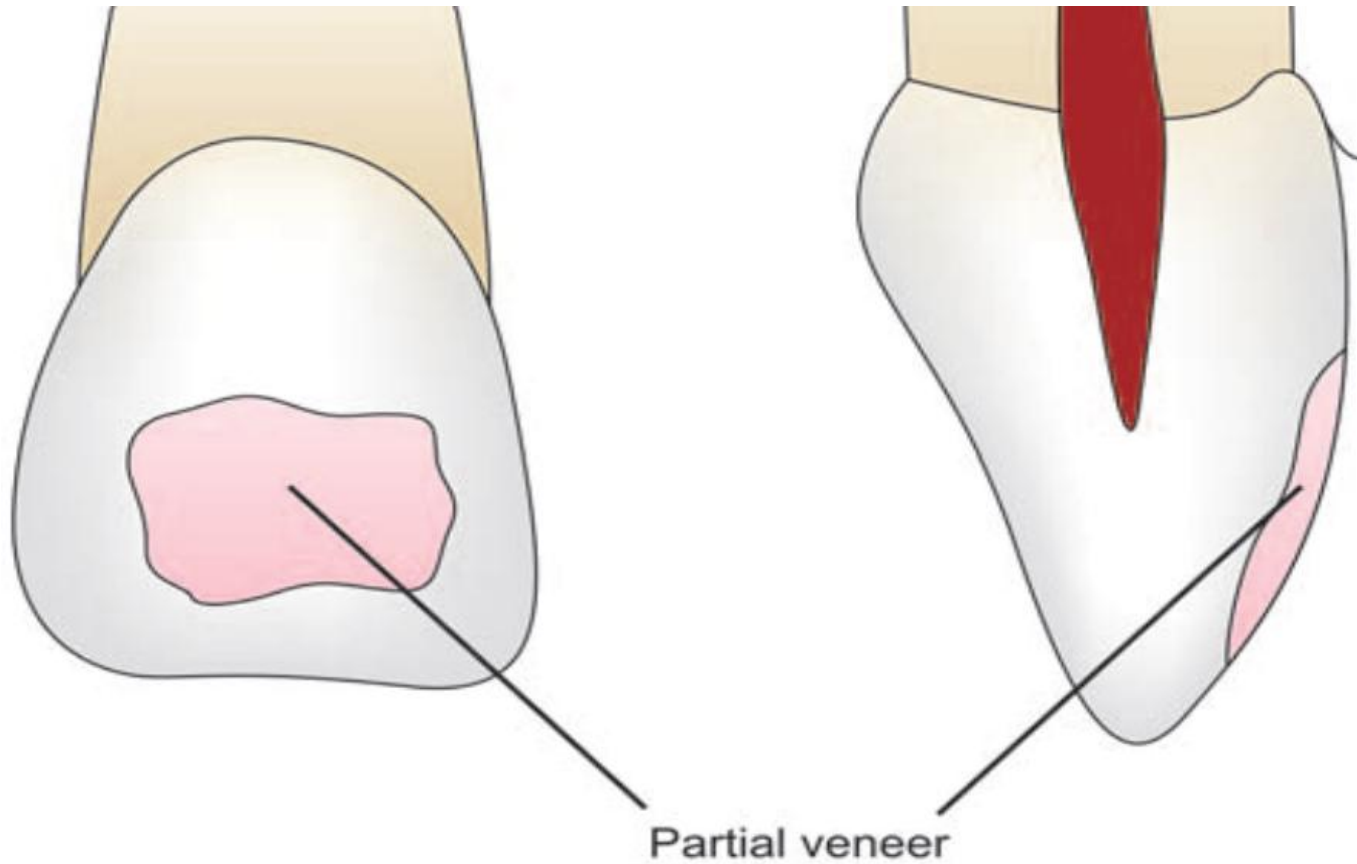
- are used when majority of facial surface or whole of the crown of a tooth is discolored.

▶ Types of Full Veneers

- Full veneer with incisal overlapping.
- Full veneer with window preparation.

Direct Veneer Technique

- ▶ Direct partial veneers are placed on localized discolorations or defects which are surrounded by sound enamel.
- ▶ Indications
 - Localized discoloration.
 - When entire facial surface is not defective.
- ▶ **Procedures**
 - Cleaning of teeth which are to be veneered.
 - Selection of the shade
 - Isolation of the teeth with cotton roll or rubber dam
 - Removal of the defect and tooth preparation. The preparation should extend on some sound enamel also.
 - Application of composite as usual, i.e. first acid etching and then application of primer and bonding agent followed by placement of composite.



Direct partial veneer involving only localized surface of labial part of the tooth



Partial veneer

► Advantages

- Single appointment.
- Useful for young patients.
- Useful for localized defects.
- Economical.

► Disadvantages

- More chair side time.
- Require more labor.

Direct Full Veneers

► Indications:

- Diastema closure
- Gross enamel hypoplasia of anterior teeth
- Grossly tetracycline stained teeth.

► *Procedure*

- Cleaning of the teeth
- Selection of the shade
- Tooth isolation and retraction of gingiva using retraction cords.
- Reduction of tooth using coarse round end diamond bur.
- Acid etching, washing and drying followed by application of bonding agent.
- Placement of composite in increments.



A, Pre operative



B, Post operative

Diastema closure by full veneer



A, Pre operative



B, Post operative

Treatment of tetracycline stains by full veneers



Management of unesthetic, pitted teeth by full veneers



Management of amelogenesis imperfecta case by full veneers

► *Advantages*

- Less technique sensitive.
- Last longer.
- Effective for multiple veneers.

► Disadvantages

- Expensive
- Require special tooth preparation.

Tooth preparation

- ▶ Tooth preparation for full veneers is of two types:
- ▶ a. Window preparation
- ▶ b. Incisal lapping preparation

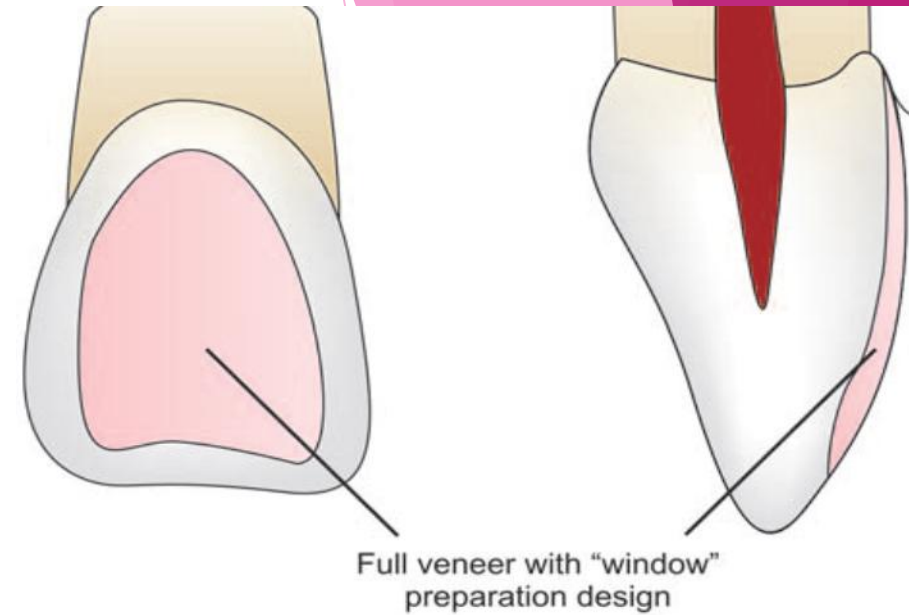
► a. Window preparation:

► Indications:

- To preserve functional lingual and incisal surfaces of anterior teeth.
- To prepare maxillary canines in patients with canine guided occlusion.

Advantages:

- Saves the functional lingual and incisal surfaces of anterior teeth.
- It does not extend subgingivally or involve incisal edge.
- Decreases the chances of wear of opposing teeth.



Full veneer with window preparation

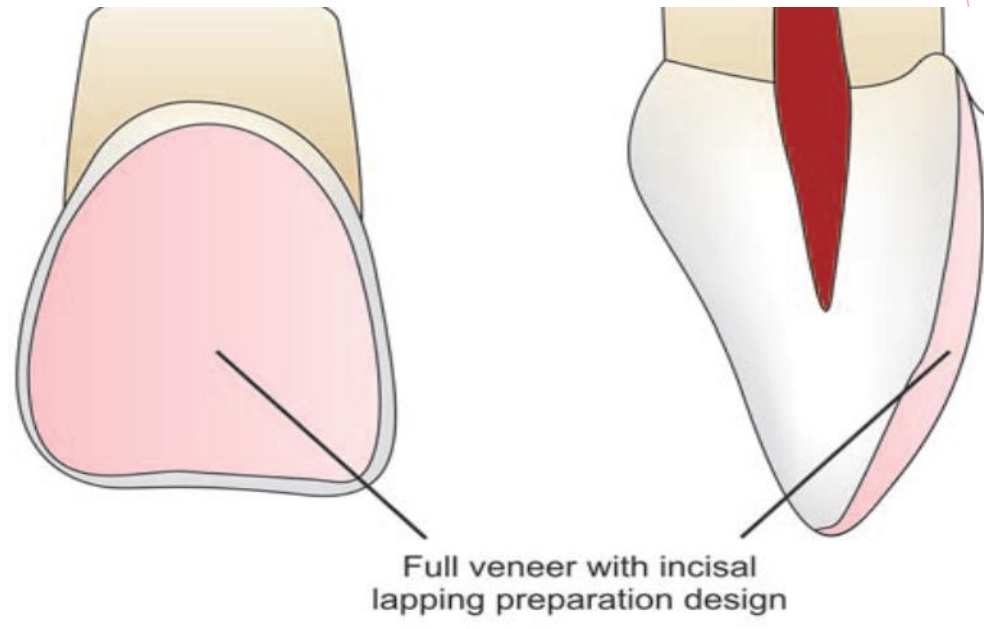
► b. Incisal lapping preparation:

► Indications:

- When crown length is to be increased.
- When incisal defect is severe and restoration is necessary.

► Advantages:

- As tooth preparation is within the enamel, hence no temporary restoration is given.
- Improved aesthetics along incisal edge.



Indirect Veneer Technique

- ▶ It is done in two appointments.
- ▶ It has following advantages:
 - Less technique sensitive than direct veneers.
 - Multiple teeth can be done in less time.
 - Chair time required for indirect veneer is less.
 - Indirect veneers produce better contour, contacts and shade.
 - These veneers have longer life than direct veneers.



► Indirect veneers are commonly made up of following:

1. Processed composite
2. Etched porcelain
3. Castable ceramic.

1. Processed Composite Veneers

► *Advantages*

- Superior physical and mechanical properties.
- Can be bonded to the teeth with a bonding agent
- Easy to finish and polish
- Can be easily repaired.
- Processed veneers are made in the cases which show attrition of anterior teeth due to occlusal stress.

Steps of veneer placement

- ▶ Processed composites have less potential to form chemical bond with bonding medium, thus additional micromechanical features are added by surface conditioning or sandblasting.
- ▶ **Acid etching** is done by using hydrofluoric acid
- ▶ After this **bonding agent** is applied to the tooth enamel.
- ▶ **Veneer is placed** by using fluid resin bonding medium.
- ▶ After **placement, finishing and polishing.**

2. Etched Porcelain Veneers

- ▶ In these porcelain veneers, internal surface is acid etched which forms stronger bond with etched surfaces of tooth.
- ▶ Advantages
 - Better retention
 - Less prone to stains
 - Good aesthetics
 - Less prone to fractures than other types of veneers.

Steps

- ▶ After cleaning and shade selection, the isolation of teeth is done.
- ▶ Tooth surfaces are prepared with round end diamond bur. Preparation should be incisal capping veneer type.
- ▶ Impression is taken with rubber base impression material and sent to laboratory for veneer formation.
- ▶ A completely finished veneer should be seated on clean, dry and isolated prepared tooth.
- ▶ Internal surfaces of porcelain veneers are conditioned with silane primer
- ▶ After setting, excess cured resin is carefully removed by BP knife.
- ▶ Re-contouring and trimming is done, if required.

3. Castable Ceramic Veneers

- ▶ Commonly used castable ceramic is 'Dicor'.
 - ▶ are fabricated for only light to moderate discolorations because it is very translucent material.
 - ▶ Formed by lost wax technique.
 - ▶ Preparation of tooth and bonding are like etched porcelain veneers.
 - ▶ are not finished with rotary instruments as rotary instruments cause loss of surface coloration.
-
- ▶ **Veneer for Metal Restoration**
 - ▶ Sometimes, veneer is placed on the facial surface of tooth which has been restored with metal.

Steps of making veneers

- ▶ Cleaning of the teeth followed by shade selection is done.
- ▶ Preparation is done by removing metal and enamel.
- ▶ Acid etching of the preparation and then drying is done.
- ▶ Placement of composite is done in usual manner.
- ▶ Finishing and polishing of the restoration is carried out.

Repair of Veneers

- ▶ Repair of veneers may be of two types:
 - ▶ 1. Veneers on tooth surface
 - ▶ 2. Veneers on the metal restoration.

- ▶ 1. Repair of veneer on tooth surface
 - a. Repair of direct composite veneers
 - ▶ - is done with the same material with which it has been prepared.
 - ▶ - After cleaning, preparation of retentive grooves and roughening the surface, composite is applied in usual manner, i.e. etching, bonding and composite placement.

b. Repair of indirect processed veneers:

i. Composite: These are repaired like direct composite veneers.

ii. Porcelain:

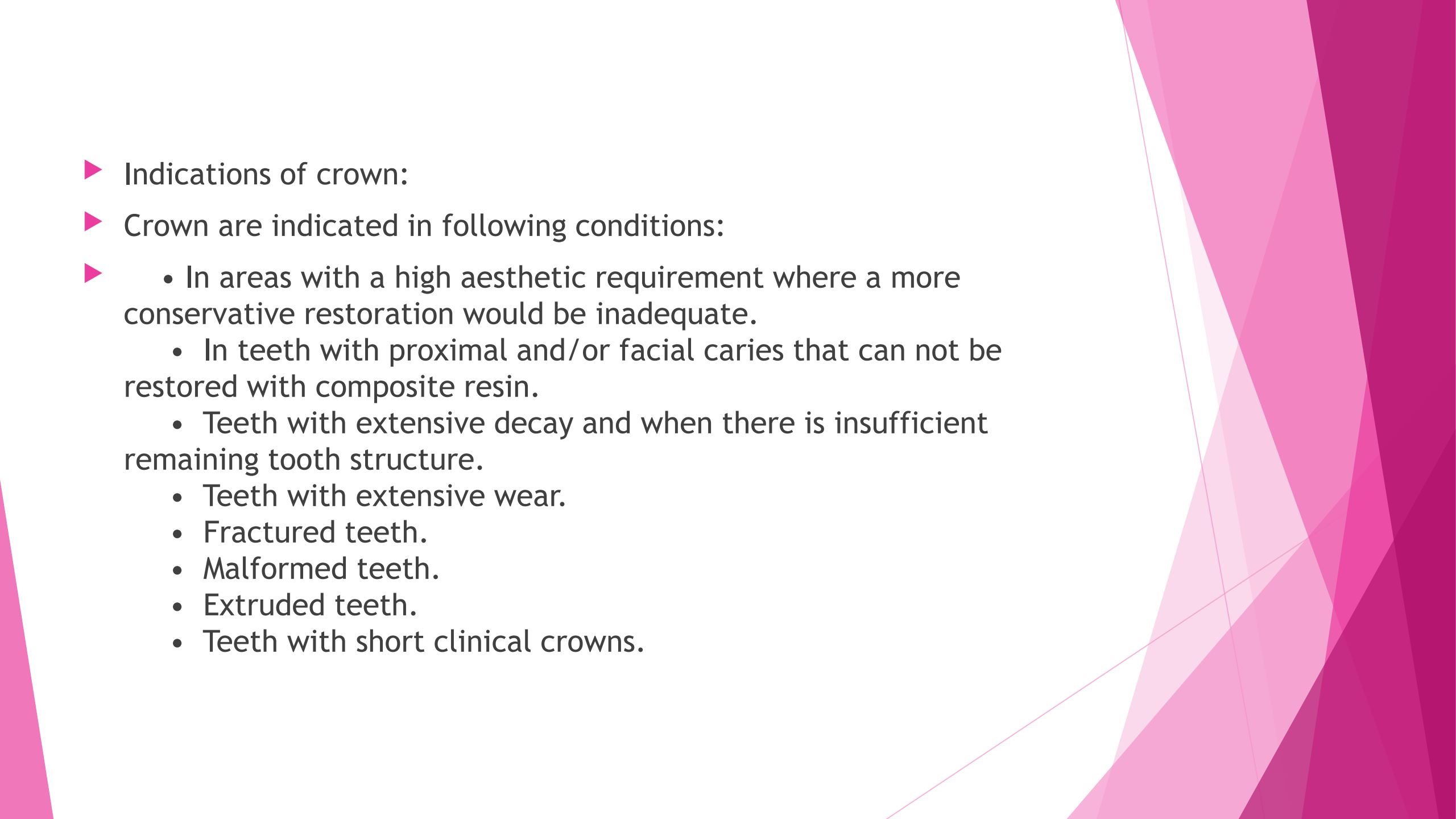
- ▶ Acid etching with 10% hydrofluoric acid (HF).
- ▶ Isolation of the tooth.
- ▶ Application of coupling agent.
- ▶ Application of resin bonding agent.
- ▶ Placement of composite, curing and then finishing.

Repair of veneer on metal restoration

- ▶ Cleaning of the tooth
- ▶ Shade selection
- ▶ Isolation of the area
- ▶ Preparation of the facial surface by removing the remaining material.
- ▶ Making chamfer finish line and retentive grooves on metal surface.
- ▶ Application of acid on the metal surface (to clean the metal surface) followed by rinsing and drying.
- ▶ Placing polyester strips and wedges interproximally to obtain proper contours.
- ▶ Application of opaque resin with a brush on the metal surface.
- ▶ For additional retention and masking effect, META is used.
- ▶ Placing and curing of composite material is done first on the cervical area in small increments followed by on entire tooth in small increments till the desired contour and aesthetic is obtained.

6. Full Coverage Crowns

- ▶ Crown
 - A crown completely covers the crown of the tooth. It can cover the tooth completely or partially. Basically full coverage restorations are required to increase the aesthetics.
 - These restorations can result in change in shape, size, contour and remarkably change self image of the patient.
- ▶ NOTE
 - In partial crown a part of the coronal surface remains uncovered and rest of the coronal tooth surface is covered like three-quarter crown and seven-eighth crown.

- 
- ▶ Indications of crown:
 - ▶ Crown are indicated in following conditions:
 - ▶
 - In areas with a high aesthetic requirement where a more conservative restoration would be inadequate.
 - In teeth with proximal and/or facial caries that can not be restored with composite resin.
 - Teeth with extensive decay and when there is insufficient remaining tooth structure.
 - Teeth with extensive wear.
 - Fractured teeth.
 - Malformed teeth.
 - Extruded teeth.
 - Teeth with short clinical crowns.

Contraindication

- ▶ A more conservative restoration can be used.
- ▶ Sufficient enamel is not present to provide adequate support.

CONCLUSION

- ▶ The contemporary concept of aesthetics is revolutionizing the way dentists diagnose, treat, and communicate with patients.
- ▶ The techniques for achieving aesthetics have improved and expanded the use of photography to analyze existing aesthetic problems and communicate possible treatment alternatives.
- ▶ Patient satisfaction is achieved when the clinician meets or exceeds the patient's expectations.
- ▶ Satisfaction is attained only through a balance in diagnosis, effective communication, and evidence-based planning and proper treatment options which can be done for the delivery of excellence in cosmetic dental treatment.

► References

- Text book of operative dentistry Jaypee brothers 2nd edition
- Sturdevant's Art and science of operative dentistry 6th edition

Thank you